Christy Li

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Professional Profile

Results-driven Web Developer with practical experience in designing and developing web applications using modern technologies such as Typescript (React, Node.js), Python, SQL, and AWS. Adapt at implementing AI models with LangChain and leveraging data analysis for predictive insights. Committed to clean, efficient coding practices and continuous learning. Seeking to contribute technical expertise and collaborative energy to a dynamic team environment.

Technical skills

- **Programming language:** JavaScript, Typescript, SQL, C#, HTML, CSS, Java, Python
- **Software framework:** React, React Native, .Net, Node.js, Express.js, Flask, Gradio, Scikit-learn, PyTorch
- **Computing Tools:** Git, Github, AWS(Lambda, SNS, S3), Firebase, Supabase, Expo, MS Office, Docker

Education Background

The University of Sydney 07/2024 – 09/2026
Master of Computer Science
Core Courses: Data Structures and Algorithms, Enterprise Scale Software Architecture, Design of Networks and Distributed Systems, Cloud Computing, Web Application Development, Object-oriented Programming
Beijing Normal - Hong Kong Baptist University 09/2018 – 06/2022
Bachelor of Business Administration (Honours) in e-Business Management and Information Systems

- **Core Courses:** Big Data Analysis and Application, Database Management, Design and Analysis of Algorithms,
- **Awards:** Second Class Award in Scholarship for the 2021-2022 academic year, 2022 MCM Honourable Mention

Volunteering and Extracurricular Activities

Technology Officer 09/2025 - Present
Skill Swap Club, The University of Sydney, New South Wales, Australia
A Skill Stack: Typescript, React, TailwindCSS, Spring Boot, Java, Supabase, Git, Agile meetings

- Developed a full-stack skill sharing platform for a university club using React, Spring Boot, and Supabase
- Designed and implemented the entire backend with Spring Boot, including REST APIs for event creation, skill posting, and a points-based system where users earn points by hosting lectures and spend points to attend events. Integrated Supabase Authentication for role-based access control. Contributed to the design of a QR code attendance feature to validate event participation. Design schema.
- Conducted code reviews for the React frontend to improve code quality, consistency, and maintainability
- Collaborated with a small team in an agile workflow, delivering features through GitHub and iterative feedback.

Web Development Project Experience

Blotz Task To-Do App 07/2025 - Present
Aim: Full-stack task management app, starting as a React Web app and later expanded to a React Native mobile app.

- Developed a full-stack Todo application using React, Next.js, and TypeScript, integrating shadcn/ui and TailwindCSS to build a modern, responsive UI and utilising Zustand for efficient global state management. Implemented a .NET Core Web API backend with SQL Server to enable real-time data retrieval and persistent todo management, enhancing user experience with dynamic UI features and theme toggling.
- Implemented core calendar-based agenda view in React Native using react-native-calendars (WeekCalendar) + FlatList, enabling users to browse tasks by day with dynamic updates. Implemented a subtasks bottomsheet to show subtask list and progress.
- Enhanced UX with an AI typing/loading animation using Lottie, providing responsive feedback during AI-driven interactions.
- Implemented a swipe-to-delete feature for task cards with Reanimated + Gesture Handler, integrated with the backend delete API and lightweight error handling (Snackbar), achieving a smooth, native-feeling interaction.
- Designed and implemented a backend-driven badge reward system, including database schema design, achievement evaluation logic, and RESTful APIs for dynamic badge assignment and user progression tracking.
- Collaborated with teammates to iterate on features and manage project through Git.

Distributed E-commerce Platform 10/2025
Aim: A microservices-based distributed e-commerce system built with Spring Boot, integrating external services (Bank, Delivery, Email) and managing distributed transactions with high availability and fault tolerance. (COMP5348 course project)

- Designed and implemented a microservices-based e-commerce platform using Spring Boot, RabbitMQ, and PostgreSQL, with separate services for Store, Bank, Delivery, and Email functionality.
- Implemented the Outbox Pattern and Saga-based compensation mechanism to handle distributed transactions and ensure data consistency, with automatic rollback for partial system failures.
- Built comprehensive fault tolerance mechanisms including retry logic, manual message acknowledgment, optimistic locking, and dead-letter queues for failed message processing.
- Followed a layered architecture with Spring Data JPA/Hibernate ORM, implementing DTOs, adapter pattern, and comprehensive logging for maintainability and debugging.

AWS Cloud-Native Image Captioning and Thumbnail Generation Platform 05/2025

Aim: Developed a scalable, high-availability image captioning and thumbnail generation web application using AWS cloud services, including EC2, Lambda, SNS, S3, and Aurora MySQL. (Cloud Computing course assignment, score: 20/20)

- Designed and deployed a responsive Flask web application running on an auto-scaling EC2 fleet behind an Application Load Balancer for high availability.
- Implemented serverless architecture with AWS Lambda and SNS to automatically generate captions via Gemini Vision API and thumbnails upon image upload.
- Managed secure storage using Amazon S3 and Aurora MySQL, integrating AWS Secrets Manager for credential protection.
- Conducted comprehensive performance testing and validation, demonstrating effective scaling and fault tolerance under high load conditions.

Professional Experience

Web Developer Internship 08/2025 – Present

BuildLess.online, Sydney, New South Wales, Australia

Skill Stack: Typescript, React, TailwindCSS, Firebase, Cloudinary, Git, Figma (UI review), Agile meetings, Playwright, Jest

- Contributed to a startup web application that enables users to build interactive UIs and logic using JSON and JavaScript stored in Firestore, with a no-code/low-code development vision.
- Implemented modular UI features to improve usability and structure.
- Developed and tested Firebase Security Rules to control access for Firestore and Storage, using the local emulator suite to validate rule behaviour through custom test cases.
- Established comprehensive testing infrastructure using Playwright for end-to-end testing and Jest for unit testing, improving code quality and reliability
- Participated in daily team meetings, collaborated with product owner and developers, and iterated features based on feedback in an agile environment.

Algorithm Engineer 09/2022 – 04/2024

Guangdong Create Environmental Technology Co., Ltd, China

Skill Stack: Python, Data Analysis, Machine Learning, Deep Learning (CNN, LSTM), Computer Vision (YOLOv5), LLM Development (LangChain, ChatGLM3)

- Built and deployed an AI-powered onboarding chatbot using LangChain and ChatGLM3, with a Gradio based front-end for interactive Q&A, automating responses to employee policy queries and reducing HR workload.
- Designed and implemented an intelligent risk assessment system using Retrieval-Augmented Generation (RAG) and LangChain, automating insurance document analysis and significantly reducing manual review time.
- Built machine learning models (e.g., XGBoost) to monitor wastewater discharge and detect excessive emissions or leakages, ensuring regulatory compliance.
- Implemented a river water quality early-warning system using Isolated Forest for anomaly detection and LSTM for predictive forecasting, improving inspection efficiency by 25%.
- Designed and implemented a behaviour detection system, a water pollution detection system usingYOLOv5, ensure enterprises maintain equipment on time and monitor water pollution occurs.
- Led data analysis workflows with Python and SQL, producing monthly environmental reports and supporting compliance audits.
- Collaborated with cross-functional teams to deliver solutions from research to deployment, ensuring robustness, safety, and regulatory compliance in production systems.

Publications

Conference Papers

- **Li, C**., Xie, M. & Wang, M. (2021). Understanding User Engagement in Online Communities during COVID-19 Pandemic: Evidence from Sentiment and Semantic Analysis on YouTube. In Proceedings of the 21st International Conference onElectronic Business. ICEB’21, Nanjing, China, December 3-7, 2021.

Forthcoming Book

- Statistical Learning Methods (Python Language)
  - ISBN: 978-7-115-
  - Status: In Progress
  - Contribution: Python introduction, Plotline visualisation, Linear Regression, Overfitting, Decision Tree, Ensemble learning, Perceptrons & Neural Network, Convolution Neural Network

Additional Information

- **Languages:** English, Cantonese, Mandarin
- **Interests:** Writing technical summaries, Basketball, Guitar
- **Github:** https://github.com/LiChenyang-GZ
- **GitBook:** https://christy-blog.gitbook.io/ctrl+s-knowledge/
- **My Personal Website:** https://lichenyang-gz.github.io/my-personal-website/
- **Soft Skills:** Growth mindset, collaborative problem-solving, time management, continuous learning

References

Available upon request.